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DIGIT APRIL 2003

hypethesis

Grantsdale chipset

■ What is it?

Grantsdale will be Intel's most significant chips et revision in almost a decade. Expected in the year 2004, it will usher a slew of innovations into the PC, such as a new graphics core, PCI Express (formerly known as 3GIO) and the NewCard expansion card format that will support wireless communications.

■ How does it work?

The most significant change will be the inclusion of PCI Express as a graphics card interface, replacing the current AGP standard. Adopters will thus be forced to invest in a new compliant graphics card, and a new motherboard. The chipset will support the LGA 775 socket, used by the Tejas processor, and will serve as a transition between the Tejas and Prescott processors—supporting both. The chipset will support both a 533 MHz and a 800 MHz front-side bus.

■ What does it do?

It supports wireless communication and offers next-generation graphics cards a significant boost in data bandwidth, over the current AGP interface. PCI Express uses a series of point-to-point channels to communicate information, in either x1, x2, x4, x8, x12, x16, or x32 dual channels, with 2.0 Gbps of available bandwidth per channel. The chipset also supports up to either DDR333 or DDR400 memory.

Discerning scanner

C current face-recognition technology works by matching images obtained from security cameras with two-dimensional pictures of known suspects that exist in a database. The problem with 2D documentation is that any change in the facial angle, lighting condition or expression can lead to the system's failure.

Russian identical twins, Alex and Michael Bronstein, claim to be staring at the face of a solution. Their technique maps the human face in three dimensions, by using algorithms and a 3D facial scanner. The surface of a face is scanned using points of light, and stored within a computer as a 3D-image. The computer uses algorithms to measure

the distance between several sample points on the facial surface. These distances are converted into straight



ILLUSTRATION: Farzana Cooper

lines in 3D space, creating a digital sculpture of the original face, which is almost as unique as a fingerprint.

Having succeeded in differentiating between the twins, the system is awaiting a patent for commercial use.

Office 2003 peeks from behind the curtains

M icrosoft has unveiled new features that will constitute Office 2003, the next version of its popular office suite. Sharing the spotlight are two new members of the family: OneNote and InfoPath.

OneNote is an electronic tabbed notebook that allows you to type, draw, scribble and directly drag and embed content from Web sites. InfoPath is meant as a tool to create XML documents—the tag-language that finds

extensive service within Office 2003, especially MS Word.

Other new features include Business Contact Manager, an Outlook add-on. The Beta 2 of Outlook also offers spam-filtering features, and a new version of Publish boasts more powerful desktop-publishing features. Microsoft also plans to add Information Rights Management technology, that lets users protect documents and email messages.

3D monitors

I n an effort to bring 3D visuals to monitors and LCD screens, a consortium of high-tech companies has been formed. 3D monitors consist of two TFT panels separated by a parallax barrier. Each eye receives a slightly different image, creating the illusion of



depth. While LCD displays of the future will display 3D images without the need for special glasses

viewing objects in the foreground can cause the eyes to shift back and forth rapidly, creating eye strain. Such related health issues, and hardware and software compliance, will form the agenda of two sub-committees of the consortium.

To run programs in 3D on these screens, users will not need special glasses or additional software. Named the 3D Consortium, it includes Sony, Sanyo, Microsoft, Kodak, NTT Data and Sharp. Sharp already sells a cellphone with a 3D screen for the NTT DoCoMo network in Japan and has a notebook that runs a 3D version of Quake.

snapshot

The cellular industry in India added **5.93 lakh** subscribers to their services last month

Source: Cellular Operators Association of India

■ 72,000 Tablet PCs shipped in the fourth quarter of last year ■ Benetton plans to track inventory with chips embedded into its creations